

flight software architecture design.

→ NASA cFs :- Core flight software.

① industry standard for learning modular space architecture.

→ F Prime (F') JPL :-

① component driven software framework.

② Ingenuity mars helicopter.

③ Website contains tutorials, architectural design docs.

→ CCSDS :- Consultative committee for Space Data Systems (CCSDS).

→ Refer EPET 400 : Spacecraft mission design {open source textbook}.

→ Understand the telemetry parameters.

① spacecraft modes:-

① Launch

② deployment

③ Commissioning: Early opⁿ phase; systems are checked

④ Nominal

⑤ safe

⑥ Maintenance

→ ^{**} Telemetry threshold values varies w^ot operational modes.

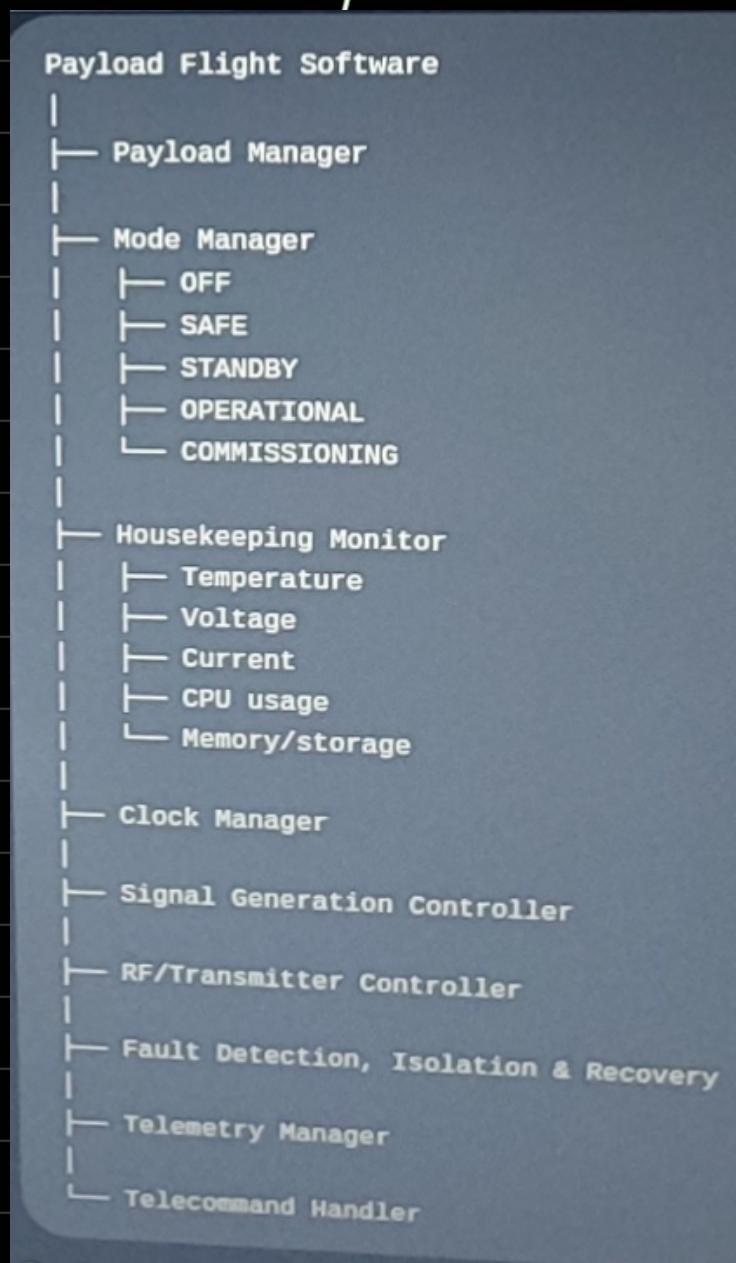
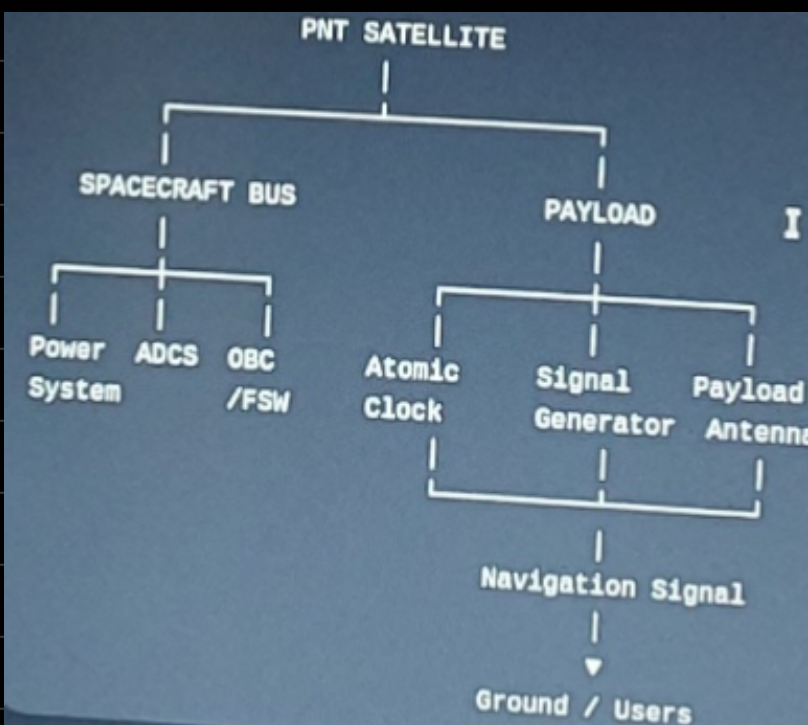
eg:- during commissioning, CPU = 90%.

is expected as engg test hardware.

But, during nominal mode, it may trigger warning.

→ Layman understanding:-

① payload = "mission equipment"
 satellite bus = "platform that supports mission op."



→ Based on my understanding; Almost all telemetry parameters will have both upper & lower threshold limits. It needs to operate within the limits. Any deviations indicates malfunction.

→ Watchdog Counter :- Mechanism that check whether software is still operating correctly.

① The specific number defined to it has no universal meaning. It could represent anything.

→ "solution type: PPP" → Precise point positioning.

→ "clock-bias-ns" → difference b/w receiver's clock and the reference time.

→ Angular-rate parameters wrt "gyro" indicate the signs of tumbling and other rotational operations.